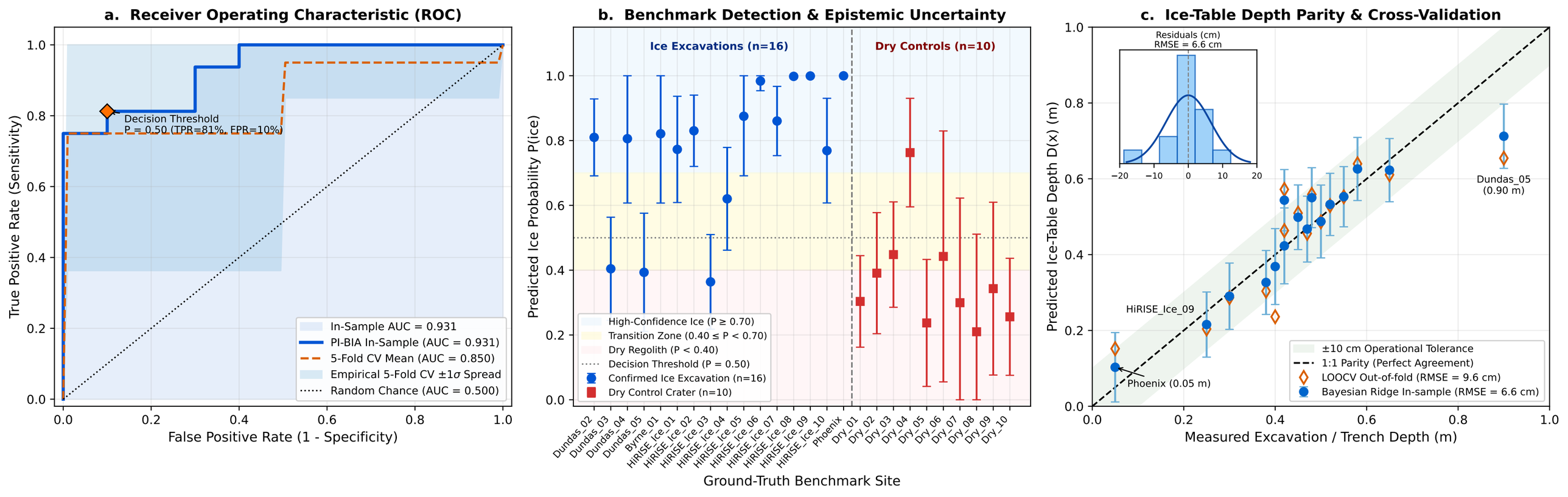
a. Receiver Operating Characteristic (ROC)
True Positive Rate (Sensitivity)
False Positive Rate (1 - Specificity)
Decision Threshold
P = 0.50 (TPR=81%, FPR=10%)
In-Sample AUC = 0.931
PI-BIA In-Sample (AUC = 0.931)
5-Fold CV Mean (AUC = 0.850)
Empirical 5-Fold CV ±1σ Spread
Random Chance (AUC = 0.500)
b. Benchmark Detection & Epistemic Uncertainty
Ice Excavations (n=16)
Dry Controls (n=10)
Predicted Ice Probability P(ice)
High-Confidence Ice (P ≥ 0.70)
Transition Zone (0.40 ≤ P < 0.70)
Dry Regolith (P < 0.40)
Decision Threshold (P = 0.50)
Confirmed Ice Excavation (n=16)
Dry Control Crater (n=10)
Dundas_02
Dundas_03
Dundas_04
Dundas_05
Byrne_01
HiRISE_Ice_01
HiRISE_Ice_02
HiRISE_Ice_03
HiRISE_Ice_04
HiRISE_Ice_05
HiRISE_Ice_06
HiRISE_Ice_07
HiRISE_Ice_08
HiRISE_Ice_09
HiRISE_Ice_10
Phoenix
Dry_01
Dry_02
Dry_03
Dry_04
Dry_05
Dry_06
Dry_07
Dry_08
Dry_09
Dry_10
Ground-Truth Benchmark Site
c. Ice-Table Depth Parity & Cross-Validation
Residuals (cm)
RMSE = 6.6 cm
Predicted Ice-Table Depth D(x) (m)
Measured Excavation / Trench Depth (m)
Dundas_05
(0.90 m)
HiRISE_Ice_09
Phoenix (0.05 m)
±10 cm Operational Tolerance
1:1 Parity (Perfect Agreement)
LOOCV Out-of-fold (RMSE = 9.6 cm)
Bayesian Ridge In-sample (RMSE = 6.6 cm)